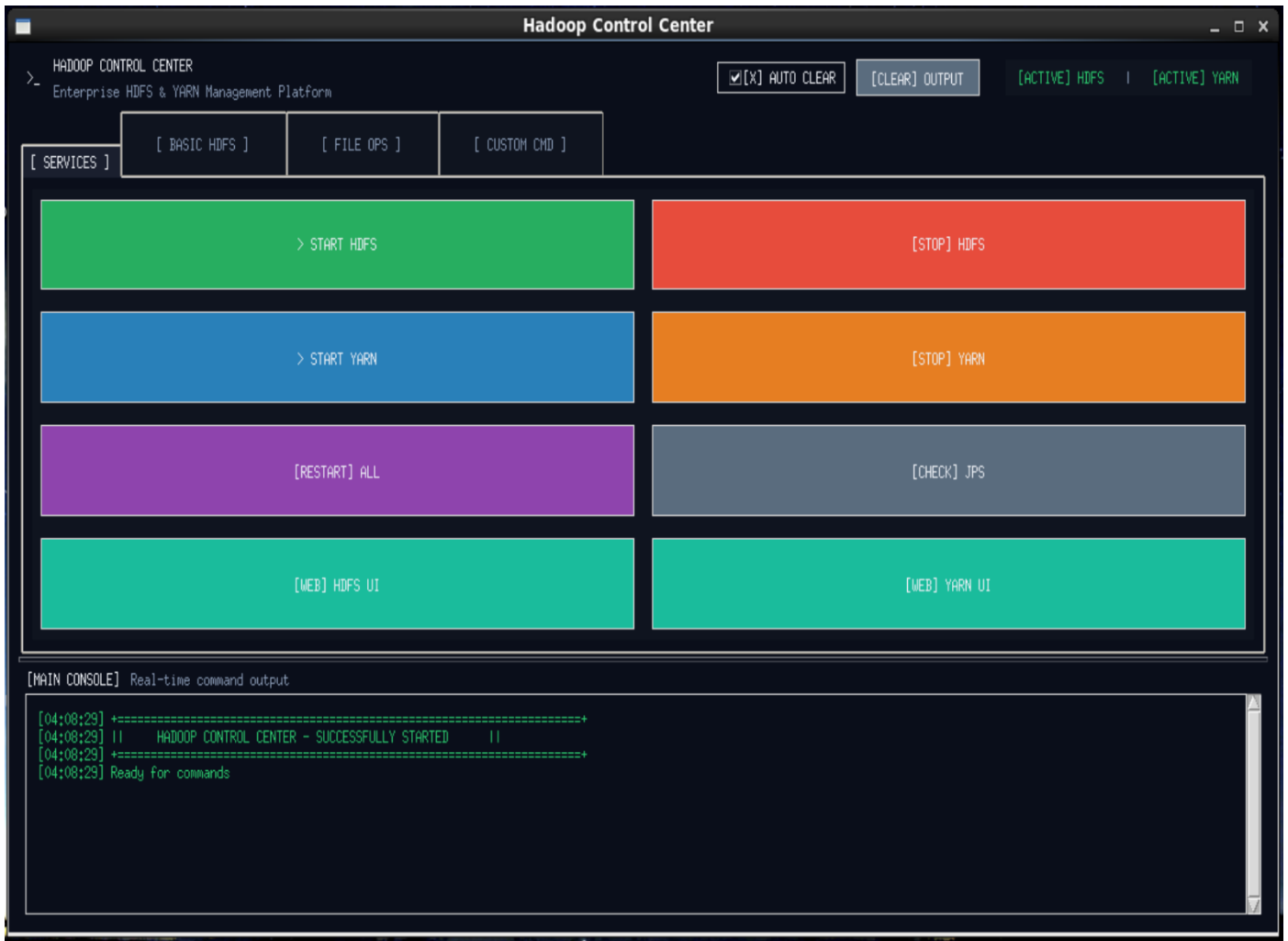


# Ali Mersal - Task 1

```
bigdata@localhost:~/Desktop
File Edit View Search Terminal Help
(base) [bigdata@localhost ~]$ cd Desktop
(base) [bigdata@localhost Desktop]$ python3 gui.py
```



cannot run multiple operations simultaneously. Please wait for the current operation to finish.

## start hdfs

The screenshot shows the Hadoop Control Center interface. At the top, the title bar reads 'Hadoop Control Center'. Below it, the subtitle is 'Enterprise HDFS & YARN Management Platform'. On the right, there are buttons for '[X] AUTO CLEAR', '[CLEAR] OUTPUT', and status indicators for '[ACTIVE] HDFS' and '[ACTIVE] YARN'. The main area is divided into two columns of buttons. The left column contains: '> START HDFS' (green), '> START YARN' (blue), '[RESTART] ALL' (purple), and '[WEB] HDFS UI' (teal). The right column contains: '[STOP] HDFS' (red), '[STOP] YARN' (orange), '[CHECK] JPS' (grey), and '[WEB] YARN UI' (teal). Below the buttons is a 'MAIN CONSOLE' section with a scrollable text area showing the following log output:

```
[04:18:15] [OK] Console cleared
[04:18:15] -----
[04:18:15] | EXECUTING: Starting HDFS Services
[04:18:15] | COMMAND: /home/bigdata/hadoop-2.7.3/sbin/start-dfs.sh
[04:18:15] -----
[04:18:15] Starting namenodes on [localhost]
[04:18:19] localhost: starting namenode, logging to /home/bigdata/hadoop-2.7.3/logs/hadoop-bigdata-namenode-localhost.localdomain.out
[04:18:24] localhost: starting datanode, logging to /home/bigdata/hadoop-2.7.3/logs/hadoop-bigdata-datanode-localhost.localdomain.out
[04:18:24] Starting secondary namenodes [0.0.0.0]
[04:18:28] 0.0.0.0: starting secondarynamenode, logging to /home/bigdata/hadoop-2.7.3/logs/hadoop-bigdata-secondarynamenode-localhost.localdomain.out
[04:18:29] [OK] Success (exit code: 0)
[04:18:29]
```

At the bottom left, the status '[READY]' is displayed.

## Stop hdfs

The screenshot shows the Hadoop Control Center interface after the 'stop hdfs' operation. The title bar and subtitle are the same. The status indicators on the right now show '[INACTIVE] HDFS' and '[ACTIVE] YARN'. The button layout is identical to the previous screenshot. The 'MAIN CONSOLE' section shows the following log output:

```
[04:16:25] [OK] Console cleared
[04:16:25] -----
[04:16:25] | EXECUTING: Stopping HDFS Services
[04:16:25] | COMMAND: /home/bigdata/hadoop-2.7.3/sbin/stop-dfs.sh
[04:16:25] -----
[04:16:26] Stopping namenodes on [localhost]
[04:16:31] localhost: stopping namenode
[04:16:36] localhost: stopping datanode
[04:16:37] Stopping secondary namenodes [0.0.0.0]
[04:16:42] 0.0.0.0: stopping secondarynamenode
[04:16:43] [OK] Success (exit code: 0)
[04:16:43]
```

## start yarn

The screenshot shows the Hadoop Control Center interface. At the top, there's a header with 'HADOOP CONTROL CENTER' and 'Enterprise HDFS & YARN Management Platform'. On the right, there are buttons for '[X] AUTO CLEAR', '[CLEAR] OUTPUT', and status indicators for '[ACTIVE] HDFS' and '[ACTIVE] YARN'. Below the header, there are three tabs: '[ SERVICES ]', '[ BASIC HDFS ]', '[ FILE OPS ]', and '[ CUSTOM CMD ]'. The '[ SERVICES ]' tab is selected. It contains a grid of buttons: '> START HDFS' (green), '[STOP] HDFS' (red), '> START YARN' (blue), '[STOP] YARN' (orange), '[RESTART] ALL' (purple), '[CHECK] JPS' (grey), '[WEB] HDFS UI' (teal), and '[WEB] YARN UI' (teal). Below the grid is a '[MAIN CONSOLE]' section with a scrollable text area showing real-time command output. The output shows the execution of the 'start-yarn.sh' script, including starting the resource manager and node manager. At the bottom left, there is a '[READY]' status indicator.

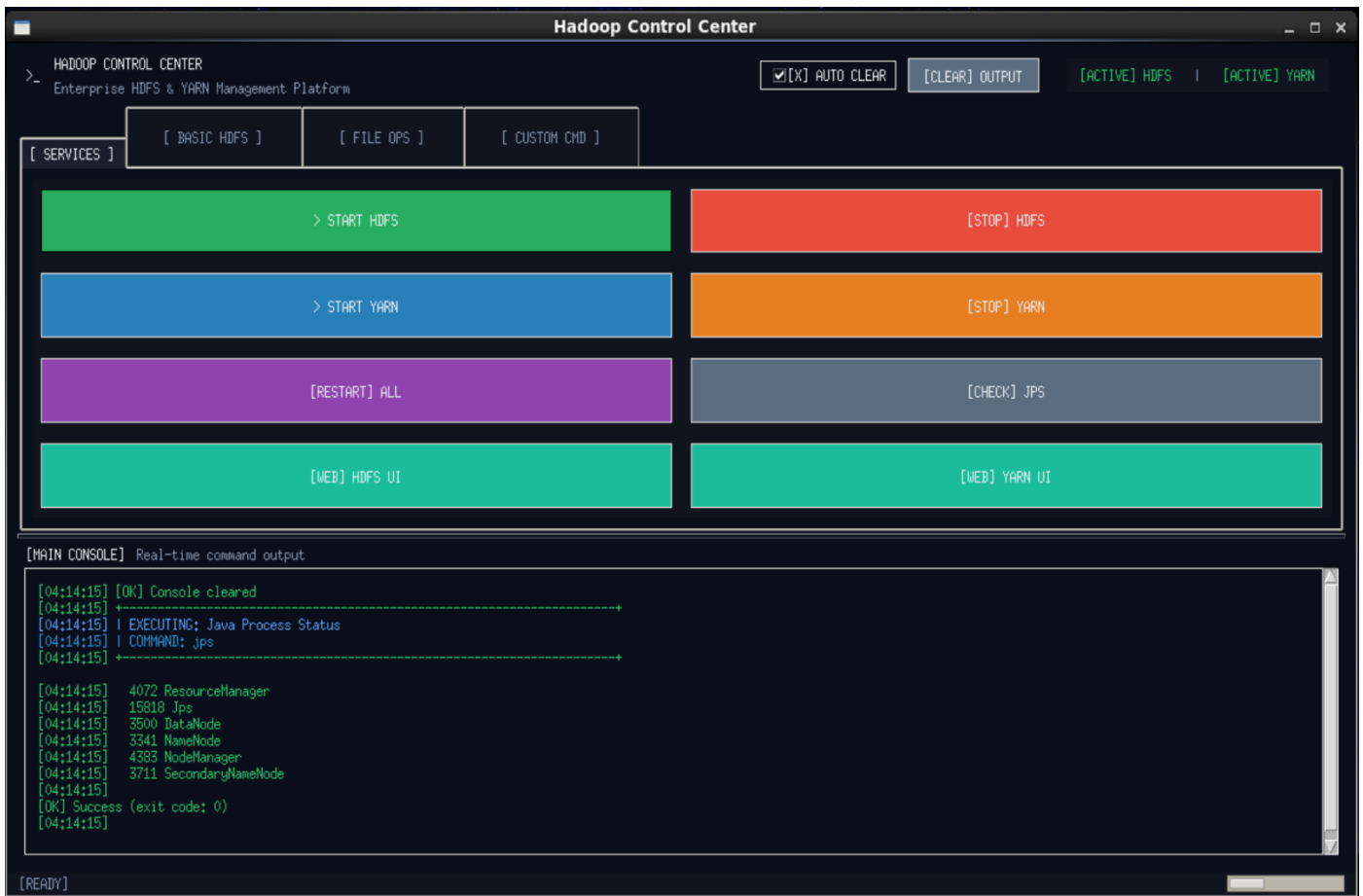
```
[04:22:01] [OK] Console cleared
[04:22:01] -----
[04:22:01] | EXECUTING: Starting YARN Services
[04:22:01] | COMMAND: /home/bigdata/hadoop-2.7.3/sbin/start-yarn.sh
[04:22:01] -----
[04:22:01] starting yarn daemons
[04:22:01] starting resourcemanager, logging to /home/bigdata/hadoop-2.7.3/logs/yarn-bigdata-resourcemanager-localhost.localdomain.out
[04:22:04] localhost: starting nodemanager, logging to /home/bigdata/hadoop-2.7.3/logs/yarn-bigdata-nodemanager-localhost.localdomain.out
[04:22:04] [OK] Success (exit code: 0)
[04:22:04]
```

## Stop yarn

The screenshot shows the Hadoop Control Center interface after the 'stop yarn' command. The status indicators at the top right now show '[ACTIVE] HDFS' and '[INACTIVE] YARN'. The '[MAIN CONSOLE]' section shows the execution of the 'stop-yarn.sh' script, including stopping the resource manager and node manager. The output shows the successful completion of the stop process. The '[READY]' status indicator remains at the bottom left.

```
[04:19:59] [OK] Console cleared
[04:19:59] -----
[04:19:59] | EXECUTING: Stopping YARN Services
[04:19:59] | COMMAND: /home/bigdata/hadoop-2.7.3/sbin/stop-yarn.sh
[04:19:59] -----
[04:19:59] stopping yarn daemons
[04:19:59] stopping resourcemanager
[04:20:10] localhost: stopping nodemanager
[04:20:10] no proxyserver to stop
[04:20:10] [OK] Success (exit code: 0)
[04:20:10]
```

# JPS

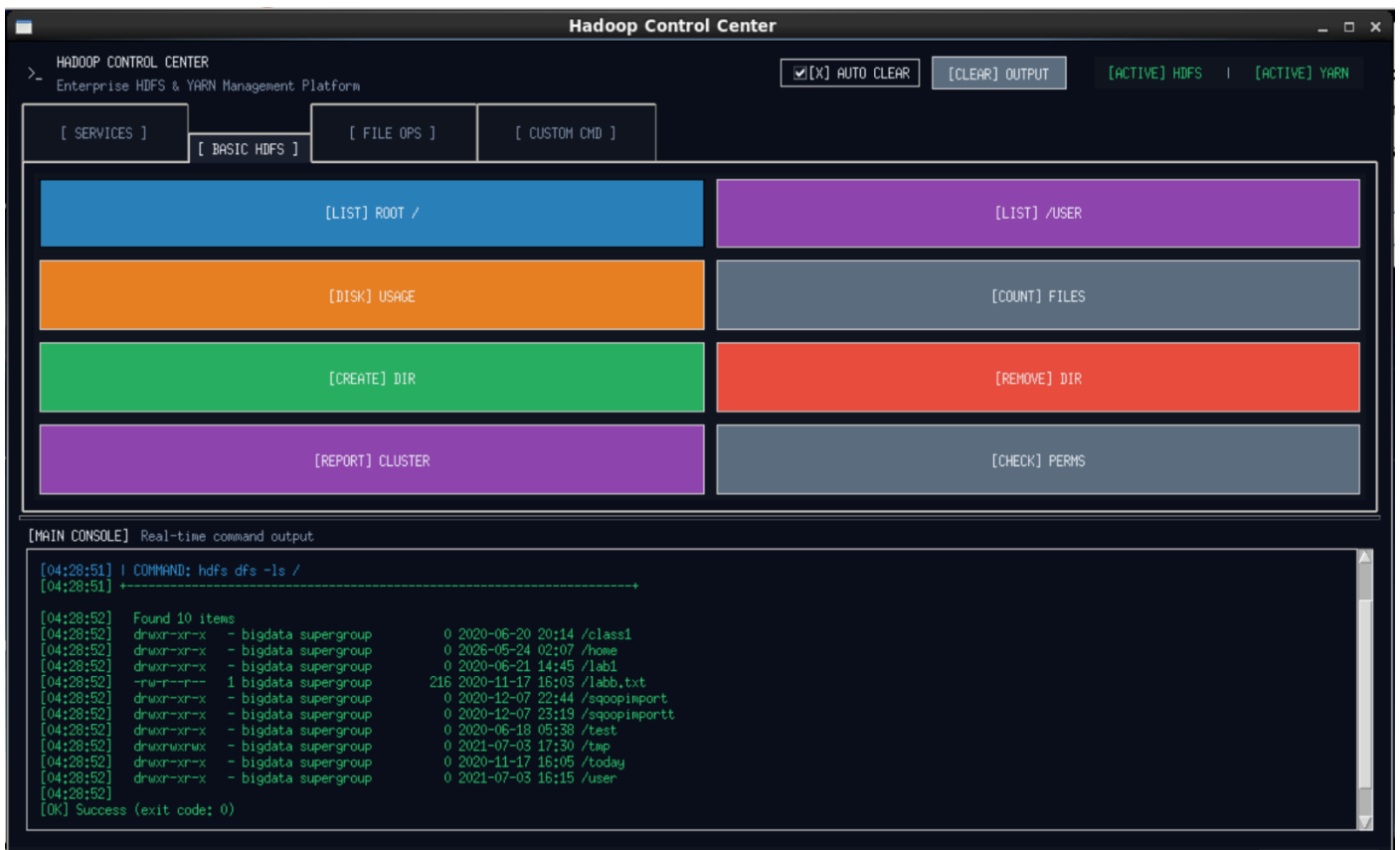


The screenshot shows the Hadoop Control Center interface with the 'JPS' (Java Process Status) tab selected. The interface includes a top navigation bar with 'HADOOP CONTROL CENTER' and 'Enterprise HDFS & YARN Management Platform'. Below this are tabs for '[ SERVICES ]', '[ BASIC HDFS ]', '[ FILE OPS ]', and '[ CUSTOM CMD ]'. The main area contains a grid of buttons: '> START HDFS', '[STOP] HDFS', '> START YARN', '[STOP] YARN', '[RESTART] ALL', '[CHECK] JPS', '[WEB] HDFS UI', and '[WEB] YARN UI'. At the bottom is a '[MAIN CONSOLE]' section showing real-time command output.

```
[04:14:15] [OK] Console cleared
[04:14:15] +-----+
[04:14:15] | EXECUTING: Java Process Status
[04:14:15] | COMMAND: jps
[04:14:15] +-----+

[04:14:15] 4072 ResourceManager
[04:14:15] 15818 Jps
[04:14:15] 3500 DataNode
[04:14:15] 3341 NameNode
[04:14:15] 4383 NodeManager
[04:14:15] 3711 SecondaryNameNode
[04:14:15]
[OK] Success (exit code: 0)
[04:14:15]
```

# LIST Root



The screenshot shows the Hadoop Control Center interface with the 'LIST Root' tab selected. The interface includes a top navigation bar with 'HADOOP CONTROL CENTER' and 'Enterprise HDFS & YARN Management Platform'. Below this are tabs for '[ SERVICES ]', '[ BASIC HDFS ]', '[ FILE OPS ]', and '[ CUSTOM CMD ]'. The main area contains a grid of buttons: '[LIST] ROOT /', '[LIST] /USER', '[DISK] USAGE', '[COUNT] FILES', '[CREATE] DIR', '[REMOVE] DIR', '[REPORT] CLUSTER', and '[CHECK] PERMS'. At the bottom is a '[MAIN CONSOLE]' section showing real-time command output.

```
[04:28:51] | COMMAND: hdfs dfs -ls /
[04:28:51] +-----+

[04:28:52] Found 10 items
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2020-06-20 20:14 /class1
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2026-05-24 02:07 /home
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2020-06-21 14:45 /lab1
[04:28:52] -rw-r--r-- 1 bigdata supergroup 216 2020-11-17 16:03 /labb.txt
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2020-12-07 22:44 /sqoopimport
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2020-12-07 23:19 /sqoopimportt
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2020-06-18 05:38 /test
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2021-07-03 17:30 /tmp
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2020-11-17 16:05 /today
[04:28:52] drwxr-xr-x - bigdata supergroup 0 2021-07-03 16:15 /user
[04:28:52]
[OK] Success (exit code: 0)
```

# Count Files

HADOOP CONTROL CENTER

Enterprise HDFS & YARN Management Platform

[X] AUTO CLEAR

CLEAR OUTPUT

ACTIVE HDFS

ACTIVE YARN

SERVICES

BASIC HDFS

FILE OPS

CUSTOM CMD

LIST ROOT /

DISK USAGE

CREATE DIR

REPORT CLUSTER

LIST /USER

COUNT FILES

REMOVE DIR

CHECK PERMS

MAIN CONSOLE

Real-time command output

[04:34:14] [OK] Console cleared

[04:34:14] EXECUTING: [COUNT] FILES

[04:34:14] COMMAND: hdfs dfs -count /

[04:34:15] 52 50 3120356 /

[04:34:16] [OK] Success (exit code: 0)

# Report Cluster

HADOOP CONTROL CENTER

Enterprise HDFS & YARN Management Platform

[X] AUTO CLEAR

CLEAR OUTPUT

ACTIVE HDFS

ACTIVE YARN

SERVICES

BASIC HDFS

FILE OPS

CUSTOM CMD

LIST ROOT /

DISK USAGE

CREATE DIR

REPORT CLUSTER

LIST /USER

COUNT FILES

REMOVE DIR

CHECK PERMS

MAIN CONSOLE

Real-time command output

[04:35:47] [OK] Console cleared

[04:35:47] EXECUTING: [REPORT] CLUSTER

[04:35:47] COMMAND: hdfs dfsadmin -report

[04:35:48] Configured Capacity: 37885648896 (35.28 GB)

[04:35:48] Present Capacity: 24548171776 (22.86 GB)

[04:35:48] DFS Remaining: 24544698368 (22.86 GB)

[04:35:48] DFS Used: 3473408 (3.31 MB)

[04:35:48] DFS Used%: 0.01%

[04:35:48] Under replicated blocks: 0

[04:35:48] Blocks with corrupt replicas: 0

[04:35:48] Missing blocks: 0

[04:35:48] Missing blocks (with replication factor 1): 0

[04:35:48] Live datanodes (1):

[04:35:48] Name: 127.0.0.1:50010 (localhost)

[04:35:48] Hostname: localhost

[04:35:48] Decommission Status : Normal

[04:35:48] Configured Capacity: 37885648896 (35.28 GB)

[04:35:48] DFS Used: 3473408 (3.31 MB)

[04:35:48] Non DFS Used: 13337477120 (12.42 GB)

[04:35:48] DFS Remaining: 24544698368 (22.86 GB)

[04:35:48] DFS Used%: 0.01%

[04:35:48] DFS Remaining%: 64.79%

[04:35:48] Configured Cache Capacity: 0 (0 B)

[04:35:48] Cache Used: 0 (0 B)

[04:35:48] Cache Remaining: 0 (0 B)

[04:35:48] Cache Used%: 100.00%

[04:35:48] Cache Remaining%: 0.00%

[04:35:48] Xcelfers: 1

[04:35:48] Last contact: Sun May 24 04:35:48 EET 2026

[04:35:49] [OK] Success (exit code: 0)

READY

# File Operation (put -get – cat -chmod-delete)

## Upload

The screenshot shows the Hadoop Control Center interface. At the top, there are tabs for [ SERVICES ], [ BASIC HDFS ], [ FILE OPS ], and [ CUSTOM CMD ]. The [ FILE OPS ] tab is selected. Below the tabs, there is a section for [PATH CONFIGURATION] with two input fields: LOCAL: /home/bigdata/Desktop/book.txtS and HDFS: /user/bigdata/. There are buttons for [BROWSE] and [VERIFY]. Below this is a section for [FILE OPERATIONS] with buttons for [UPLOAD], [DOWNLOAD], [VIEW], [CHMOD], and [DELETE]. The [UPLOAD] button is highlighted. At the bottom, there is a [MAIN CONSOLE] section showing real-time command output. The output shows the command 'hdfs dfs -put "/home/bigdata/Desktop/book.txt" "/user/bigdata/book.txt"' being executed successfully.

```
[04:49:57] [OK] Console cleared
[04:49:57] +-----+
[04:49:57] | EXECUTING: Upload: book.txt
[04:49:57] | COMMAND: hdfs dfs -put "/home/bigdata/Desktop/book.txt" "/user/bigdata/book.txt"
[04:49:57] +-----+
[04:49:59]
[OK] Success (exit code: 0)
[04:49:59]
```

The screenshot shows the Hadoop Browse Directory page. The address bar shows 'localhost:50070/explorer.html#/user/bigdata'. The page has a green header with navigation links: Hadoop, Overview, Datanodes, Snapshot, Startup Progress, and Utilities. The main content area is titled 'Browse Directory'. Below the title is a search bar with the text '/user/bigdata' and a 'Go!' button. Below the search bar is a table showing the contents of the directory.

| Permission | Owner   | Group      | Size | Last Modified                   | Replication | Block Size | Name     |
|------------|---------|------------|------|---------------------------------|-------------|------------|----------|
| drwx-----  | bigdata | supergroup | 0 B  | Sun 24 May 2026 05:00:00 AM EET | 0           | 0 B        | .Trash   |
| -rw-r--r-- | bigdata | supergroup | 77 B | Sun 24 May 2026 04:49:58 AM EET | 1           | 128 MB     | book.txt |

# View

HADOOP CONTROL CENTER

Enterprise HDFS & YARN Management Platform

☒[X] AUTO CLEAR

[CLEAR] OUTPUT

[ACTIVE] HDFS

[ACTIVE] YARN

[ SERVICES ]

[ BASIC HDFS ]

[ FILE OPS ]

[ CUSTOM CMD ]

[PATH CONFIGURATION]

LOCAL:

[BROWSE]

HDFS:

[VERIFY]

[FILE OPERATIONS]

[UPLOAD]

[DOWNLOAD]

[VIEW]

[CHMOD]

[DELETE]

[MAIN CONSOLE] Real-time command output

```
[04:51:16] [OK] Console cleared
[04:51:16] -----
[04:51:16] | EXECUTING: View: /user/bigdata/book.txt
[04:51:16] | COMMAND: hdfs dfs -cat "/user/bigdata/book.txt"
[04:51:16] -----
[04:51:17] read the book pls
[04:51:17] -----
[04:51:17] -----
[04:51:17] -----
```

# Chmod

HADOOP CONTROL CENTER

Enterprise HDFS & YARN Management Platform

☒[X] AUTO CLEAR

[CLEAR] OUTPUT

[ACTIVE] HDFS

[ACTIVE] YARN

[ SERVICES ]

[ BASIC HDFS ]

[ FILE OPS ]

[ CUSTOM CMD ]

[PATH CONFIGURATION]

LOCAL:

[BROWSE]

HDFS:

[VERIFY]

[FILE OPERATIONS]

[UPLOAD]

[DOWNLOAD]

[VIEW]

[CHMOD]

[DELETE]

[MAIN CONSOLE] Real-time command output

```
[05:00:55] =====
[05:00:55] || HADOOP CONTROL CENTER - SUCCESSFULLY STARTED ||
[05:00:55] =====
[05:00:55] Ready for commands
[05:01:40] [OK] Selected: /home/bigdata/Desktop/book.txt
[05:01:42] [ERROR] HDFS path required
[05:01:44] [ERROR] Enter HDFS path First
```

Change Permissions

CHANGE PERMISSIONS

Path: /user/bigdata/book.txt

Mode (e.g., 755):

[APPLY]

[CANCEL]

# Custom Cmd

## HADOOP CONTROL CENTER

Enterprise HDFS & YARN Management Platform

☒ [X] AUTO CLEAR

[CLEAR] OUTPUT

[ACTIVE] HDFS

[ACTIVE] YARN

[ SERVICES ]

[ BASIC HDFS ]

[ FILE OPS ]

[ CUSTOM CMD ]

[CUSTOM HADOOP COMMAND]

hdfs dfs -text /user/bigdata/mapreduce/word.txt | head -50

[EXECUTE] (Ctrl+Enter)

[COMMAND EXAMPLES]

hdfs dfs -mkdir -p /user/bigdata/input

-> Create directory

hdfs dfs -text /user/bigdata/mapreduce/word.txt | head -50

-> View first 50 lines

hdfs dfs -count -q -h /user

-> Show quota

hdfs dfsadmin -report

-> Cluster report

[MAIN CONSOLE] Real-time command output

[05:10:41] [OK] Console cleared

[05:10:41] |-----|

[05:10:41] | EXECUTING: Custom Command

[05:10:41] | COMMAND: hdfs dfs -text /user/bigdata/mapreduce/word.txt | head -50

[05:10:41] |-----|

[05:10:42] Hello World Bye World

[05:10:42] Hello Hadoop Bye Hadoop

[05:10:42] Bye Hadoop Hello Hadoop

[05:10:42]